Mohammad Saim

saimmd@mail.uc.edu | thisissaim | Google Scholar | Cincinnati, OH

Research Interests

Affective Computing · Social Computing · Computational Cognitive Science · Large Language Models

Education

PhD in Computer Science & Engineering *Aug 2024 – Present*
University of Cincinnati GPA: 3.88 / 4.0
Advisor: Dr. Tianyu Jiang *Focus: Language Modeling for Emotion Understanding*

Bachelor of Technology in Computer Engineering *2020 – 2024*
Aligarh Muslim University GPA: 9.07 / 10.0

Research & Industry Experience

Doctoral Research | *University of Cincinnati* *Aug 2024 – Present*

- **Emotion perception:** Developed **ELENA**, a zero-shot LVLM framework for embodied emotion recognition (up to +15.6 F1 over baselines); built **CHIARO**, a 1,000-scene appraisal-grounded benchmark on which the strongest LLM reaches only 67.3 macro-F1, well below human agreement.
- **Emotion in social reasoning:** Showed that induced emotions reverse LLM moral decisions in up to 20% of cases; built a 500-scenario emotional labor dataset revealing that persona-conditioned LLMs favor deep acting, driven by Conscientiousness and Emotional Stability.

Graduate Research Assistant | *Procter & Gamble, Digital Accelerator* *Aug 2026 – Present*

- Apply sequential learning to consumer product formulation data (hair and beauty care), pairing surrogate modeling with Bayesian optimization to steer ingredient selection toward target performance.
- Build an interactive UI module that lets product testers log results and review model-recommended formulations, closing the experiment loop.

Graduate Research Assistant | *University of Cincinnati (Sponsored: General Electric)* *May 2026 – Aug 2026*

- Developed a rule-based design management system for 2D and 3D CAD models, integrating custom-trained embeddings, OCR-based drawing extraction, and a VLM chatbot to automate manufacturability verification.

Summer Research Intern | *IIT Jodhpur, School of AI & Data Science* *May 2024 – Jul 2024*
Supervised by Prof. Ganesh Manjhi

- Performed geospatial analysis of water surface area changes in Jharkhand using satellite imagery, WRIS datasets, OpenStreetMap, and QGIS; produced temporal analysis of reservoir and non-reservoir water body variation.

Select Publications

[1] **Saim M.**, Jiang T. "Emotional Labor Strategy Preferences in LLM Personas." *To appear in Findings of the 2026 Conference on Empirical Methods in Natural Language Processing* **(EMNLP 2026)**.

[2] Bommana D., **Saim M.**, Jiang T. "Chiaroscuro for Emotions: A Contrastive Emotion Benchmark Grounded in Appraisal Theory." *To appear in the 2026 Conference on Empirical Methods in Natural Language Processing* **(EMNLP 2026)**.

[3] **Saim M.**, Jiang T. "Do Emotions Influence Moral Judgment in Large Language Models?" *Findings of the 64th Annual Meeting of the Association for Computational Linguistics* **(ACL 2026)**.

[4] **Saim M.**, Duong P. A., Luong C., Bhanderi A., Jiang T. "Anatomy of a Feeling: Narrating Embodied Emotions via Large Vision-Language Models." *Findings of the 2025 Conference on Empirical Methods in Natural Language Processing* **(EMNLP 2025)**.

[5] **Saim M.**, Rizvi R., Khan M. "Analyzing the Performance of ML and DL Models in Detecting Cyberbullying Comments." *ICRASET 2023*, pp. 1–6.

[6] **Saim M.**, Rizvi R. M. "Digital Resilience: A Review of Cutting-Edge Approaches to Cyberbullying Detection in the Social Media Landscape." *IDICAIEI 2023*, pp. 1–6.

One additional first-author manuscript under double-blind review in NAACL 2026; details available upon request.

Teaching Experience

Graduate Teaching Assistant | *University of Cincinnati* *Aug 2024 – Present*

- **Formal Languages & Automata** (Spring '26, Dr. Guan) – Assisted students with finite automata, context-free grammars, and pumping lemma proofs.
- **Operating Systems** (Fall '25, Prof. Gallagher) – Supported kernel compilation, module development, and custom Linux kernel build assignments.
- **Intelligent Data Analysis** (Spring '25, Prof. Atluri) – Guided students through frequent pattern mining, classification algorithms, and statistical modeling techniques.
- **Machine Learning** (Fall '24, Prof. Ralescu) – Covered regression, Bayesian learning, gradient ascent in ANNs, and SMO; assisted with grading and doubt clearance.

Technical Skills

Languages	Python, C, C++, JavaScript, Solidity
ML/DL	PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, Keras, OpenCV
LLMs & Foundation Models	Gemini/Claude APIs, PEFT, RAG, vector DBs, LLM-as-judge, TransformerLens
Optimization	Bayesian optimization, Gaussian process surrogates, sequential experimental design
Web & Viz	React, Flask, Streamlit, Gradio, Matplotlib, Seaborn, Tableau
Tools	Git, Docker, LaTeX, QGIS, Adobe Suite

Honors & Awards

- **Best PhD Student Award (2025–26)**, Department of Computer Science & Engineering, University of Cincinnati
- **IEEE Lance Stafford Larson Student Award 2023** – recognized for excellence in technical content, writing, and presentation
- **Sir Syed Mentoring & Assistance Fellowship** (2023–24)
- **Third Rank**, Engineering & Technology proposals, In-House Students' Research Convention, AMU Innovation Council [News]

Select Projects

Comment Toxicity Classifier *[]* *2023*

- Built a multi-label toxic comment classifier using BERT, DistilBERT, and GRU, achieving over 90% accuracy; deployed via a Gradio interface.

Real-Time Object Detection for Visually Impaired | *Prof. Sarosh Umar* *2023*

- Developed an assistive mobile application using MobileNet SSD with voice feedback, OCR, and haptic navigation for blind users.

Symposia & Conference Presentations

- **64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)**, San Diego, USA (Poster).
- **2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)**, Suzhou, China (Poster).
- **Asian Undergraduate Symposium (2024)**, NUS – Presented a plan to revive low-resource languages. (Poster)
- International Conference on Recent Advances in Science & Engineering Technology (ICRASET), 2023.
- Intl. Conference on Computing, Communication & Networking Technologies (ICCCNT), IIT Delhi, 2023.
- International Conference on Futuristic Technologies (INCOFT), Belgaum, 2022.

Leadership & Service

Chairperson, IEEE Computer Society | *Zakir Husain College of Engineering & Technology* *2023–24*

- Organized three workshops on ML/AI attended by 60+ students; led editorial team for the society's inaugural annual newsletter; oversaw inter-club coding competitions.

Academic Guide | *SSMAF Fellowship* *2025–Present*

- Mentor fellowship students through study-abroad applications, from program selection to statements of purpose.

Academic Service

- **Reviewer**, Conference on Neural Information Processing Systems (NeurIPS 2026).
- **Reviewer**, ACL Rolling Review (ARR), March 2026 cycle.
- **Industry Judge**, College of Engineering and Applied Science (CEAS) Expo, University of Cincinnati, 2026.
- **Reviewer**, LREC-COLING 2025 International Conference on Language Resources and Evaluation.